

CHE 2201 – Physical Chemistry II

Quantum Mechanics

Lec. 1. Birth of Quantum Mechanics

1.1. Introduction

Quantum Mechanics is the physics of matter at much smaller scales than we are able to observe or feel. As a result, we have no direct experience of this area of physics and therefore no intuition of how such microscopic systems behave. Since the mass and dimensions of microparticles are extremely small compared with those of macroscopic bodies, the properties and regularities of the motion of a microparticle differ significantly from those of the motion of a macroscopic body, which have been described by Classical (or Newtonian) Mechanics.

Classical Mechanics (or Classical Physics) was the result of the scientific revolution of the 16th and 17th centuries, and the core of this physics is Newton's laws describing the motion of particles of matter. The particles are subjected to forces, and Newton's second law ($F = ma$) can be used to describe the motion of the particle. By specifying the position and velocity of the particle at some initial time, the motion of the particle could be determined at a subsequent time. In classical mechanics, there is nothing that prevents the initial conditions from being determined with certain accuracy. After Newton, the major direction of physics was to incorporate as much as possible the physical phenomena into the framework of Newtonian Mechanics.

One such important application was the description of wave motion in fluids and elastic materials. Since the waves are seen as deviations of some quantity such as the height of the surface of water from some average value, the wave is characterized by the amplitude of this deviation and the sign of the deviation at any point and time may be either positive or negative. Since the waves are solutions to linear differential equations, waves can be added by simply adding the deviation of the wave from equilibrium at every point at any given time. The fact that this deviation may be either positive or negative leads to the wave motion being either cancelled or enhanced at different points which produces

the typical wave phenomena of interference and diffraction. The average energy density carried by a mechanical wave is proportional to the square of the amplitude of the wave, and is independent of the frequency.

At the beginning of the 19th century, one of the most prominent problems was the question of the nature of light. The long and heated argument as to whether light is a stream of particles or a wave, was finally settled in the 19th century, by demonstrating in experiments that light could interfere and diffract, and was therefore a wave.

During the course of this century, the understanding electromagnetic phenomena developed rapidly, finally resulting in Maxwell's equations for electromagnetic fields. In this case, the great achievement was the prediction of wave solutions to Maxwell's equations that led to the unification of electrodynamics and optics. The Maxwell's equations were also verified by the discovery of radio waves by Hertz. But, there were still obstacles to the unification of electromagnetism with mechanics. The nature of electromagnetic currents was not understood until the very last part of the century and there appeared to be no supporting medium for electromagnetic waves, as it was the case with all mechanical waves. An attempt to deal with the latter problem was to propose the existence of a universal medium called the 'ether', which was assumed to be the medium that supported electromagnetic waves. However, the unsatisfactory nature of this hypothesis led Einstein to develop the special theory of relativity in 1905.

The situation with thermodynamics was much more satisfactory. The discovery that heat was a form of energy that could be produced from mechanical energy (or work) and that could be in turn used to perform work (including mechanical work), led to the concept of conservation and conversion of energy. That is, energy can be transformed from one type to another but cannot be created or destroyed. During the 19th century, the laws and the structure of thermodynamics were formulated and applied to a variety of phenomena. In addition, through the kinetic theory of gasses and the development of statistical mechanics, it was shown that thermodynamics could be described by the average motions of complicated systems of very large numbers of particles. Since the exact motion of such

large collections of particles could not be determined, statistical methods are used to describe average properties of macroscopic systems.

While studying the background of the development of quantum mechanics, it is important to note that the 19th century is also the time when chemistry became a quantitative science. It was discovered that the light spectrum given by various materials when heated, showed discrete lines rather than a continuous distribution of wavelengths. The periodic table of elements (Mendeleev, 1869) showed that the chemical properties of various elements display a regular pattern as a function of the atomic number. As the century progressed, more elements were discovered and the improved spectroscopic information about these elements was accumulated. However, the new experimental data accumulated were not fully incorporated into the framework of Newtonian Mechanics.

Let us now consider few examples of experimental findings that led to the development of quantum mechanics.

1.2. Black Body Radiation

One of the phenomena that had been known for a long time was that when a metal is heated to a sufficiently high temperature, it begins to glow and the colour of the glowing metal object depends on the temperature. This was used by metal smiths (workers) to determine when metal was sufficiently hot to be easily worked with hammers. Here, the scientific explanation is that a hot object emits electromagnetic radiation. At high temperatures, an appreciable proportion of the radiation is in the visible region of the spectrum, and a higher proportion of short-wavelength blue light is generated as the temperature is raised. In 1859, on the basis of thermodynamics, Kirchhoff showed that the energy density (the energy per unit area or volume) of light emitted by a completely absorbing body is a function of the temperature of the object and the frequency of the emitted light.

An object that is capable of absorbing all the light (i.e. electromagnetic radiation) falling on it, is referred to as a Black Body. More precisely, a Black Body is an ideal system capable of absorbing and emitting all wavelengths of radiation uniformly. A good

approximation to a black body is an empty container with a pinhole, maintained at a constant temperature (see Fig. 1.1). In such a case, any radiation going through the hole will be absorbed and re-emitted inside so many times as it is reflected around inside the container.

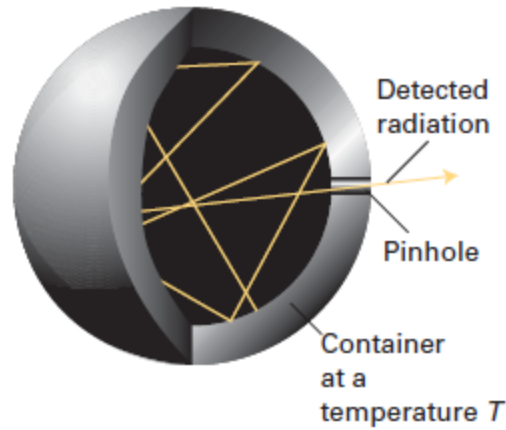


Fig. 1.1. Experimental Representation of a Black Body

The approach adopted by 19th century scientists to explain black body radiation was to calculate the Energy Density. It is the total energy in a region of the electromagnetic field divided by the volume of the region (units: J m⁻³), due to all the oscillations corresponding to wavelengths between λ and $\lambda + d\lambda$.

This energy density dE is proportional to the width of this range of wavelengths $d\lambda$, and is written as

$$dE = \rho \times d\lambda \quad (1.1),$$

where $\rho(\lambda, T)$, the proportionality constant between dE and $d\lambda$, is called the Density of States (units: J m⁻⁴).

A high density of states at the wavelength λ and temperature T , simply means that there is a lot of energy associated with wavelengths lying between λ and $(\lambda + d\lambda)$ at that temperature.

The total energy density in a region is the integral over all wavelengths:

$$E(T) = \int_0^{\infty} \rho(\lambda, T) d\lambda \quad (1.2)$$

The classical model of black body radiation, which is known as the Rayleigh-Jeans law for the Density of States, is as follows:

$$\rho(\lambda, T) = 8\pi kT/\lambda^4 \quad (1.3),$$

where k is Boltzmann's constant ($k = 1.381 \times 10^{-23} \text{ J K}^{-1}$).

Although the Rayleigh-Jeans law is quite successful at long wavelengths (low frequencies), it fails badly at short wavelengths (high frequencies). Thus, as λ decreases, ρ increases without going through a maximum.

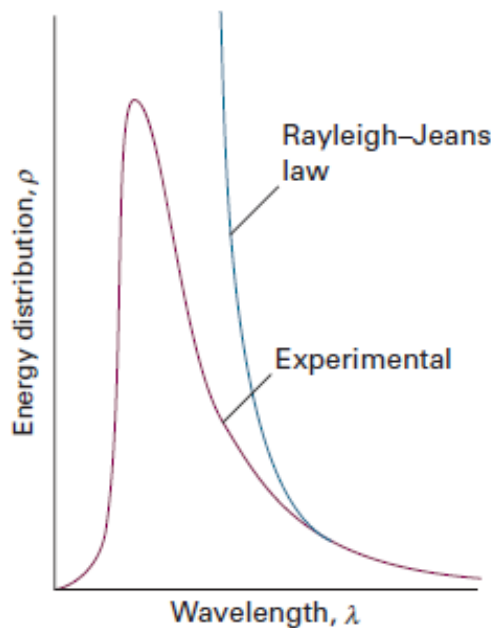


Fig. 1.2. Rayleigh-Jeans Law for the Density of States

So, the equation 1.3 predicts that oscillations of very short wavelengths (corresponding to ultraviolet radiation, X-rays, and even γ -rays) are strongly excited even at room temperature. The total energy density in a region (i.e. the integral in equation 1.2), is also predicted to be infinite at all temperatures above zero. This absurd result, which implies that a large amount of energy is radiated in the high-frequency region of the electromagnetic spectrum, is called the ultraviolet catastrophe. According to classical physics, even cool objects should radiate in the visible and ultraviolet regions, so objects should glow in the dark. It means that, in fact, there should not be any darkness.

In 1900, the German physicist Max Planck found that he could account for these experimental observations by assuming that the energy of each electromagnetic oscillation is limited to discrete values and cannot be varied arbitrarily. This suggestion is contrary to the viewpoint of classical physics, in which all possible energies are allowed and every oscillator has a mean energy kT .

The limitation of energies to discrete values is called the quantization of energy, which states that the “*energy is not propagated and transmitted, absorbed or emitted continuously but discretely, in separate portions called quanta.*” (Refer lecture on Quantum Mechanical Model of the Atom under CHE 1201 – General Chemistry).

In particular, Planck found that he could account for the observed distribution of energy if it is assumed that the permitted energies E of an electromagnetic oscillator of frequency ν are integer multiples of $h\nu$.

$$E = nh\nu \quad (1.4),$$

where $n = 0, 1, 2, \dots$;

and h is a fundamental constant, known as Planck’s constant ($h = 6.63 \times 10^{-34}$ J sec).

On the basis of this assumption, Planck was able to derive the Planck distribution:

$$\rho(\lambda, T) = \frac{8\pi hc}{\lambda^5 (e^{\frac{hc}{\lambda kT}} - 1)} \quad (1.5)$$

This expression fits the experimental curve very well at all wavelengths. It is easy to see why Planck’s approach was successful while Rayleigh’s was not. The thermal motion of the atoms in the walls of the black body excites the oscillators of the electromagnetic field. According to classical mechanics, all the oscillators of the field share equally in the energy supplied by the walls, so even the highest frequencies are excited. The excitation of very high frequency oscillators results in the ultraviolet catastrophe. But, according to Planck’s hypothesis (i.e. quantization of energy), oscillators are excited only if they can acquire an energy of at least $h\nu$. This energy is too large for the walls to supply in the case of the very high frequency oscillators, so the latter remain unexcited. The effect of

quantization is to reduce the contribution from the high frequency oscillators, for they cannot be significantly excited with the energy available.

1.3. Atomic and Molecular Emission Spectra

Another clear evidence for the quantization of energy came from spectroscopy, the detection and analysis of the electromagnetic radiation absorbed or emitted by a substance. The record of light intensity transmitted by an atom (or a molecule) as a function of frequency ν or wavelength λ is called its spectrum (from the Latin word for appearance). The emission spectrum of hydrogen atom in the visible region is given in the figure 1.3, and an obvious feature of this spectrum is that it is a line (discrete) spectrum and not a continuous one. (Please refer to lecture on Atomic Spectrum of Hydrogen under CHE 1201 – General Chemistry).

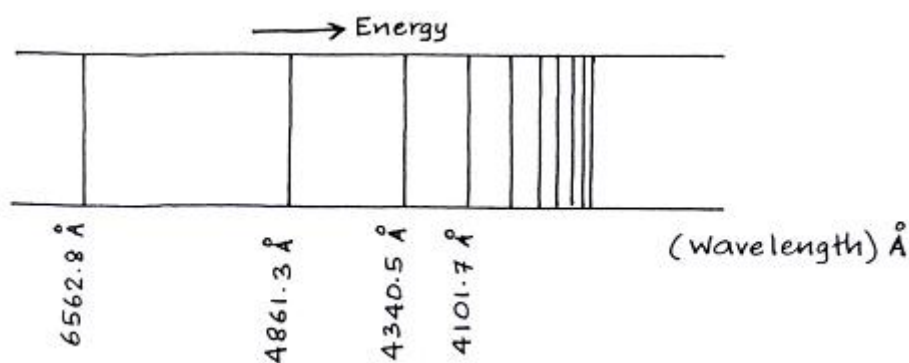


Fig. 1.3. Emission Spectrum of Hydrogen Atom in the Visible Region

This observation can be understood if the energy of the atoms (or molecules) is also confined to discrete values, for the energy can be absorbed or emitted only in discrete amounts. Then, if the energy of an atom decreases by ΔE , the corresponding energy is emitted as radiation of frequency ν , and an emission 'line' appears in the spectrum.

We say that an atom (or a molecule) undergoes a spectroscopic transition (i.e. a change of state), and the corresponding energy change ΔE can be written as follows.

$$\Delta E = E_2 - E_1 = h\nu \quad (1.6)$$

where E_2 and E_1 are the energies of the two energy levels involved.

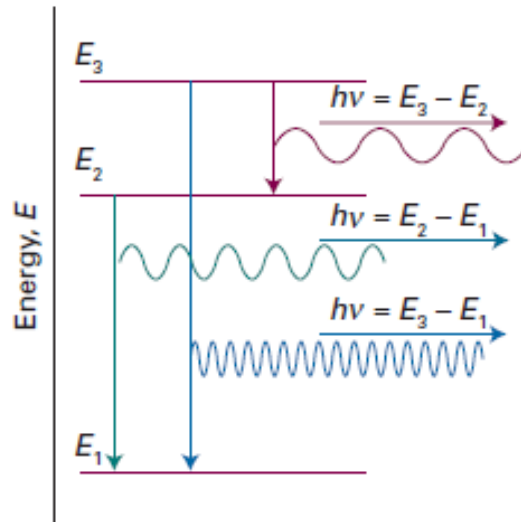


Fig. 1.4. Spectroscopic Transitions between Discrete Energy Levels

And, the frequency of emitted radiation is

$$\nu = \Delta E/h = (E_2 - E_1)/h \quad (1.7)$$

Since E_2 and E_1 can have only discrete values, the frequency of the emitted radiation (ν) can also have only certain discrete values. It has to be noted that this idea of discrete energy levels was impossible to incorporate into the framework of Newtonian Mechanics.

1.4. Photo-Electric Effect

One experiment showed that electromagnetic radiation, which classical physics treats as wave-like, also displays the characteristics of particles. Another experiment showed that photons (and electrons), which classical physics treats as particles, also display the characteristics of waves. The experimental evidence for the existence of photons came from the measurement of the energies of electrons produced in the Photoelectric Effect.

The photoelectric effect is the emission of electrons by certain metals when they are radiated with visible light, or in some cases ultraviolet light. The experimental characteristics of the photoelectric effect are as follows.

1. No electrons are ejected unless the frequency of the radiation exceeds a certain value, characteristic of the metal.

2. The kinetic energy of the ejected electrons increases linearly with the frequency of the radiation, but is independent of the intensity of the radiation.
3. Even at low light intensities, electrons are ejected immediately if the frequency is above the threshold.

These observations strongly suggests that the photoelectric effect depends on the ejection of an electron, when it is involved in a collision with an object (particle?) that carries sufficient energy to eject the electron from the metal. If we suppose that object is a photon of energy $h\nu$ (where ν is the frequency of the radiation), then the conservation of energy requires that the kinetic energy of the ejected electron ($\frac{1}{2} m_e v^2$) should obey the following equation.

$$\frac{1}{2} m_e v^2 = h\nu - \Phi \quad (1.8)$$

In this expression Φ is a characteristic of the metal called its' Work Function. It is the energy required to remove an electron from the metal to infinity (the analogue of the ionization energy of an individual atom or a molecule).

We can now see that the existence of photons accounts for the three observations we have summarized:

1. Photo-ejection cannot occur if $h\nu < \Phi$, because the photon does not bring sufficient energy.
2. Equation 1.8 predicts that the kinetic energy of an ejected electron should increase linearly with frequency.
3. When a photon collides with an electron, it gives up all its energy, so we should expect electrons to appear as soon as the collisions begin, provided the photons have sufficient energy.

1.5. Birth of Quantum Mechanics

So, it is clear that the classical mechanics (or classical physics) has failed to explain many experimental findings of that period such as black body radiation, photo electric effect,

atomic and molecular emission spectra etc. In other words, these new experimental findings could not be incorporated into the framework of classical mechanics, and the scientists had to look for new concepts. As a result, a new branch of physics, namely Quantum Mechanics, arose in the 1920s, which describes the motion and interaction of microparticles. Here, we should note that quantum mechanics can be applied to the motion of all particles, large and small, microscopic and macroscopic bodies. However, when quantum mechanics is applied to bodies which are much larger than atoms, the results obtained are identical to those obtained by using the Newton's laws of motion. However, the equations of quantum mechanics are much more complicated than those used in Newtonian mechanics, and therefore, we still use the Newton's laws to describe the motion of macroscopic bodies. It is only when we deal with microparticles; more difficult equations of quantum mechanics have to be solved.

The quantum mechanics is based on the concepts of a quantum of energy, the dual nature of the motion of microparticles, and the probability (statistical) method of describing micro-objects.